

RESEARCH ARTICLE | FEBRUARY 08 2024

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Dung N. Pham ; Sathwik Bharadwaj ; L. R. Ram-Mohan  



J. Appl. Phys. 135, 064401 (2024)

<https://doi.org/10.1063/5.0187988>



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Cite as: J. Appl. Phys. 135, 064401 (2024); doi: 10.1063/5.0187988

Submitted: 17 November 2023 · Accepted: 16 January 2024 ·

Published Online: 8 February 2024



Dung N. Pham,^{1,2,a)} Sathwik Bharadwaj,^{1,2} and L. R. Ram-Mohan^{2,3,b)}

AFFILIATIONS

¹Department of Physics, Worcester Polytechnic Institute, Worcester, Massachusetts 01609, USA

²Center for Computational NanoScience, Worcester Polytechnic Institute, Worcester, Massachusetts 01609, USA

³Departments of Physics, Electrical and Computer Engineering, and Mechanical Engineering, Worcester Polytechnic Institute, Worcester, Massachusetts 01609, USA

^{a)}**Present address:** Department of Electrical and Computer Engineering, Princeton University, Princeton, NJ 08540, USA.

^{b)}**Author to whom correspondence should be addressed:** LRRAM@wpi.edu

ABSTRACT

Semiconductor quantum dots (QDs) are one of the leading candidates for realizable qubits, as well as for many other advances in quantum computing and quantum communication. The spatial overlapping of wavefunctions describing each single electron in these nanoscale devices results in tunable spatial entanglement. In this article, we explore the case of two electrons in two-dimensional double quantum dot systems. We compute the two-particle wavefunction through a variational method combined with Hermite finite elements and study the spatial entanglement of electrons. We show that symmetry in the geometry of the double quantum dots plays a role in obtaining optimal entanglement, while a broken symmetry can lead to additional resonances in entanglement that are associated with the crossings of states. We also show that one can finely tune the level of spatial entanglement by altering the geometry of the quantum dots or by applying external fields, which corresponds to an “entanglement spectroscopy.” Finally, we study how impurities in the potential profile of the QDs affect the level of entanglement.

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I. INTRODUCTION

Recent advances in fabricating and designing quantum dots (QDs)^{1–3} have made it possible to precisely control individual particles in artificial atoms. Consequently, there has been intensive interest, both theoretically and experimentally, in the measurement and evaluation of quantum entanglement of particles in QD systems.^{4–11} In addition to the well-studied spin entanglement, there also exists *spatial entanglement*, which arises when the particles are trapped in QDs that are close enough to each other, allowing overlaps of wavefunctions. Their spatial correlation then becomes significant in applications. This spatial entanglement can be highly sensitive to changes in the geometry and symmetry of the QDs, as well as to external perturbations.

There have been proposals to use semiconductor QDs for quantum information and quantum computing purposes.^{12–18} A thorough understanding of the spatial entanglement of locally confined electrons is crucial for realizing solid state quantum computing and quantum information devices that rely on electrons as qubits.

Earlier investigations on spatial entanglement of electrons include the study of entanglement in helium and helium-like atoms,^{19,20} as well as ground state entanglement in simplified models such as Hooke’s-atom model²¹ and one-dimensional quantum wells with point-contact Coulomb interactions.²² A study of electron entanglement in nanowires was previously done by us.²³ However, a detailed study of electron entanglement in realistic, complex QD systems in higher dimensions has not been considered. More specifically, how the level of entanglement between the particles depends on the potential profile of the system, how to design QDs to obtain certain entanglement properties, or how to finely tune the entanglement through external parameters, is of importance and has not been discussed.

In this article, we compute the spatial entanglement in double quantum dot systems and propose an entanglement spectroscopic technique to manipulate the entanglement. Two-dimensional quantum dots can be realized by embedding a quantum wire in a semiconductor material²⁴ or by using appropriate gate geometries.²⁵ We study in detail the case of two electrons trapped in

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two-dimensional double quantum dots and investigate the dependence of entanglement on the (a) QDs geometry, (b) symmetry, as well as (c) when an external magnetic field is applied. We show that spatial entanglement is maximum when the double QD system has a mirror symmetry, and the entanglement value reduces when the symmetry is broken. We also report on the presence of entanglement resonances associated with level crossings of quantum states and the formation of an electron cluster when the electron congruence occurs. Furthermore, we discuss how impurities affect the measure of entanglement of the system and how one can engineer the QDs to obtain stable entanglement constants.

We emphasize that while we are presenting results for quantum dots of simple geometrical shapes, our use of the finite element methods (FEMs) will allow us to transcend such geometrical aspects to calculate results for dots of any shape. Our choice of square, rectangular, and triangular shapes for the wells leads to a simpler understanding of the entanglement phenomena.

This paper is organized as follows: In Sec. II, we explain our approach to computing the few-particle spatial wavefunctions and quantifying the spatial entanglement. In Sec. III A, we present the results on entanglement of electrons in a double QD system with asymmetric sizes, while in Sec. III B, results on entanglement in QD systems with non-symmetric length are discussed. In Sec. III C, the effect of an external magnetic field on entanglement is discussed. In Sec. III D, we present the results for how impurities in the QD system degrade the level of entanglement, and in Sec. III E, we discuss entanglement in a system of two QDs connected via a potential tunnel. Finally, in Sec. III F, we study how the relative orientation between the two QDs affects the entanglement value in the context of double triangular quantum dots and double square quantum dots with different orientations. Concluding remarks are presented in Sec. IV.

II. THEORETICAL CONSIDERATIONS

We are interested in solving for the two-electron spatial wavefunction $\psi(\mathbf{r}_1, \mathbf{r}_2)$ that obeys the time-independent Schrödinger's equation

$$\left\{ -\frac{\hbar^2}{2} \left[\nabla_1^2 \left(\frac{1}{m_1^*} \nabla_1 \right) + \nabla_2^2 \left(\frac{1}{m_2^*} \nabla_2 \right) \right] + V(\mathbf{r}_1, \mathbf{r}_2) \right\} \psi = E\psi, \quad (1)$$

where $m_1^* \equiv m_1^*(\mathbf{r}_1)$, $m_2^* \equiv m_2^*(\mathbf{r}_2)$ are the effective masses of the two electrons, and $V(\mathbf{r}_1, \mathbf{r}_2)$ is the effective potential. $V(\mathbf{r}_1, \mathbf{r}_2)$ contains the Coulomb interaction between the electrons and also terms arising from the geometrical confinement of the system. The finite square well potentials considered here are not separable, hence estimates based on the infinite well potential or assumptions of separability will deliver incorrect eigenvalues. The FEM accounts for this non-separability very well.²⁶ The action integral that corresponds to Eq. (1) is given by

$$A = T \int d\mathbf{r}_1 d\mathbf{r}_2 \left[\frac{\hbar^2}{2} \left(\nabla_1 \psi^* \cdot \frac{1}{m_1^*} \nabla_1 \psi + \nabla_2 \psi^* \cdot \frac{1}{m_2^*} \nabla_2 \psi \right) + \psi^* (V(\mathbf{r}_1, \mathbf{r}_2) - E)\psi \right], \quad (2)$$

where T is the temporal integral over the range $[0, T]$ of our time-independent (steady state) equation. In our method, the action integral in Eq. (2) is treated using finite element analysis,²⁶ in which $\psi(\mathbf{r}_1, \mathbf{r}_2)$ is written as a linear combination of products of single-particle Hermite interpolation polynomials $N_\nu(\mathbf{r}_i)$, where $i = \{1, 2\}$ is the particle index, and ν is the polynomial index. Through minimization of the action integral, the variational coefficients are found and the wavefunction is accurately computed using the Hermite interpolation polynomials. A detailed discussion of our method of computing few-particle wavefunctions in one-dimensional structures can be found in Ref. 23. Here, we have generalized the method to two dimensions, in which the interpolation polynomials are C^2 -continuous at the nodes and C^1 -continuous across the edges of the finite element mesh. We note that to obtain high accuracy for the wavefunction of a two-electron system, special attention is needed in the evaluation of the Coulomb interaction term since it diverges at $\mathbf{r}_1 = \mathbf{r}_2$. To tackle this, we evaluate the contribution of the Coulomb term to the action integral using the Gauss quadratures method with a different Gauss order assigned to the parameter space of each electron. The one-dimensional Coulomb potential is then guaranteed to be finite during numerical calculations, and a fast convergence rate has been demonstrated.²³ This is because the Gauss points and the corresponding weights for different orders of Gauss-Legendre quadrature do not coincide. In this article, we have extended the Gauss quadrature formulation to two dimensions to efficiently compute the Coulomb potential in 2D quantum dots with an accuracy of 10^{-8} . We would also like to note that since the Gauss quadrature method allows for the exact computation of integrals that involve only polynomials, the evaluation of all other terms in the discretized action besides the Coulomb interaction is exact.

The spatial entanglement is quantified through the linear entropy,^{7,20,21,27,28} given by

$$\mathcal{E}_\ell = 1 - \text{Tr}(\rho_1^2), \quad (3)$$

where ρ_1 is the reduced density matrix, which can be written in a separable form

$$\rho = \rho_{\text{spatial}} \otimes \rho_{\text{spin}}, \quad (4)$$

where ρ_{spatial} and ρ_{spin} are the density matrices of the spatial and spin components, respectively. In this article, we focus on the spatial part of electron entanglement, while the spin component are found to contribute to the overall value of $\text{Tr}(\rho_1^2)$ by a constant scaling factor.²³ Therefore, ρ_{spin} is laid aside from our consideration. In Secs. III A–III F, we present our results on entanglement, as defined in Eq. (3), and show how the level of spatial entanglement depends on the potential configuration of the system, as well as on applied external fields.

III. RESULTS

A. Double quantum dots with asymmetric sizes

Consider a system of two electrons in the conduction band of a GaAs/Ga_xAl_{1-x}As two-dimensional double square quantum dot. For a stoichiometric ratio $x = 0.37$, the potential depth is

$V = 0.276$ eV, the effective mass of the electron $m_m^* = 0.0665 m_e$ inside, and $m_{out}^* = 0.0858 m_e$ outside the QDs.²⁹ Here, m_e is the rest mass of an electron. These values are used in all the calculations discussed below.

Here, we study how asymmetry in the potential of the QDs affects entanglement. We consider a system of two square quantum dots of sizes w_1 and w_2 , as shown schematically in the inset in Fig. 1(a). The distance between the QDs is held at $d = 10$ nm, the size of the first QD is fixed at $w_1 = 15$ nm, while w_2 is varied. The dependence of the level of entanglement on w_2 is shown in Figs. 1(a) and 1(b) for the first few states. When $w_2 = 15$ nm, the potential has a mirror symmetry, and we observe resonances in the entanglement values for some of the excited states. As shown in Fig. 1(a), for a mirror-symmetric double-QD system, the spatial entanglement of the first few low-lying states is either ≈ 0.75 or ≈ 0.5 . This is because, for the ground state and the first few excited states, each electron is well confined within its respective QD. The tunneling rate and Coulomb interaction are small enough so that the wavefunctions can be approximated by their asymptotic value when two QDs are infinitely far from each other. The two-electron spatial entanglement in such cases can be analytically shown to be either of the two values mentioned, depending on whether the two-particle state is asymptotically characterized by one or two single-particle quantum numbers. This can be intuitively explained as followed: consider a single electron in a symmetric double QD potential in which the two QDs are sufficiently far from each other such that their respective wavefunctions do not overlap. Each eigenstate of the electron is characterized by two quantum numbers n_1 and n_2 , one for each QD. Now, if there are two electrons in this system, they will repel each other so that in the steady state, each electron occupies a separate QD. Since the electrons are indistinguishable, the quantum dots are identical, the probabilities in predicting the single-particle states and the locations of these electrons are either $(1/2)^2$ when $n_1 = n_2$ or $(1/2)^3$ when $n_1 \neq n_2$, depending on whether there is an additional uncertainty in the single-particle quantum numbers. This results in the entanglement of 0.5 and 0.75, respectively, as is evident in Fig. 1(a) at $w_2 = 15$ nm = w_1 . For completeness, an analytical derivation of these asymptotic spatial entanglement values is provided in the Appendix.

As w_2 moves away from 15 nm and the system becomes asymmetric, we observe in these states, a collapse in the level of entanglement to ≈ 0.5 . This is because QDs are no longer identical, leading to the removal of ambiguity in assigning single-particle quantum numbers.²³ The amount of entanglement in the system is, therefore, reduced.

The exceptions to this collapsing of the entanglement level are the ground state and the sixth excited states. These states are (asymptotically) described by only one single-particle quantum number, hence there is no ambiguity in their constructions. Therefore, they maintain an entanglement value close to 0.5 throughout a wide range of w_2 .

Aside from the reduction of entanglement due to asymmetry, we also observe resonances in entanglement due to level crossings between adjacent states. These are the peaks R_{23} in Fig. 1(a) and R_{67} , R_{78} , R_{89} , and R_{910} in Fig. 1(b). Each of these resonances, generally labeled R_{mn} , is associated with the exchange of properties

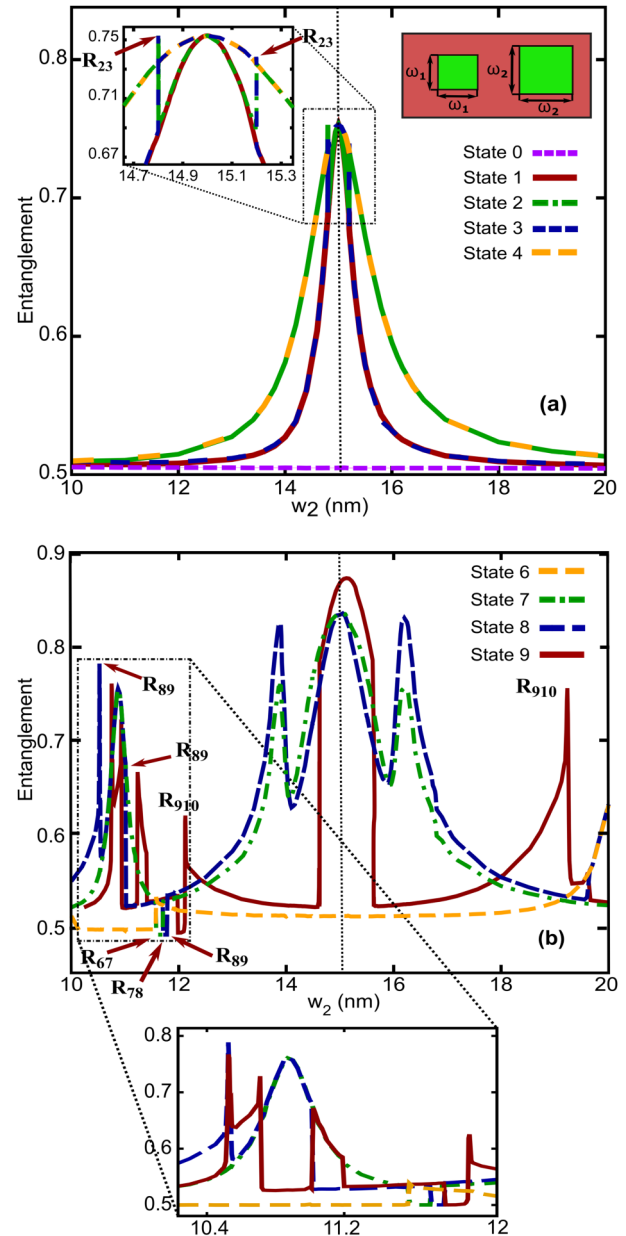


FIG. 1. Spatial entanglement of the first few eigenstates of two electrons in a double QD of non-symmetric sizes is plotted as a function of the size of the second QD w_2 . The size of the first QD is fixed at $w_1 = 15$ nm. The distance between the QDs is $d = 10$ nm. The entanglement resonances are labeled R_{pq} , where $\{p, q\}$ are the two adjacent states undergoing (anti-)crossing of states at that location.

between the m th and n th excited states. To confirm the occurrences of level crossings associated with the resonances, in Fig. 2 the electron densities of the second and third excited states before and after their shared resonance peak R_{23} are plotted. As shown in

Fig. 2, the two states exchange their properties and evolve into one another through entanglement resonance.

We note that there are certain symmetries in the evolution of the entanglement value as w_2 varies. In Figs. 1(a) and 1(b), the shapes of the entanglement plots are somewhat symmetric across the 15 nm mark. This similarity in how the entanglement responds to the increases and decreases of w_2 is due to the fact that although the size of QD2 is varied, its shape remains a square. The symmetry of each single-particle state inside this QD is, therefore, preserved, which leads to the mirror symmetry in entanglement plots in Fig. 1. We see in Fig. 1(b) that this symmetry is broken as w_2 becomes too small. This is when certain single-particle excited states in QD2 become unbounded. As w_2 keeps on decreasing, these unbound states interact with bound states and introduce additional crossings. This is evident on the left end of Fig. 1(b) and is magnified in its inset.

B. The double quantum dot with asymmetric lengths

In this section, we introduce asymmetry into the system by varying only the length ℓ_2 of QD2 while holding its width at $w_2 = w_1 = 15$ nm, effectively creating a system of a square QD and a rectangular QD. The distance between the QDs is held at $d = 10$ nm. The change in entanglement value as ℓ_2 varies is shown in Fig. 3. With the system no longer maintaining the symmetry of two square QDs as in Sec. III A, we observe an abundance of broken mirror symmetries in the entanglement value across the $\ell_2 = 15$ nm mark. This can be seen in the entanglement plots for the first, second, third, and fifth excited states in Fig. 3. The breaking of mirror symmetry is best seen in the saturation limits: In some of these states, the entanglement saturates toward two different values, 0.5 and 0.75, as it approaches the left and right ends of

the plot. An example (see Fig. 3) of such changes in entanglement limits is the case of the first excited state, which saturates at 0.75 toward the left end, and 0.5 toward the right end, or vice versa in the case of the fifth state.

Each of these shifts in the asymptotic entanglement values occurs due to level crossing between two adjacent states, during which states exchange their quantum numbers. This dynamics is shown in Fig. 3, where we observe a crossing between the second and third excited states at the symmetry mark, labeled by R_{23} , as well as the crossing of the first and second excited states that occurs nearby.

We also observe that the convergence of entanglement to asymptotic values is fast, which allows for the designing of a stable single-level entanglement system. This is illustrated in Fig. 4, where the entanglement of the third excited state as a function of ℓ_2 is shown as an example. The entanglement evolves like a step function, with the step-up location at the symmetry mark, where the crossing of this state with the second excited state occurs and the two state swap their asymptotic entanglement values. Therefore, one can build a double QD system with QD1 being a square, and depending solely on the relative lengths of two QDs to have an entangled system with a 0.75 ($\ell_2 > \ell_1$) entanglement, or 0.5 ($\ell_2 < \ell_1$). The amount of entanglement will remain unperturbed, given that the relative conditions are maintained.

The breaking of mirror symmetry in entanglement, along with the shifting of asymptotic entanglement values was not observed previously in 1D systems, since they lack the freedom in distorting the shape of the potential confinements used to trap the electrons; stretching or squeezing a 1D quantum well or quantum wire does

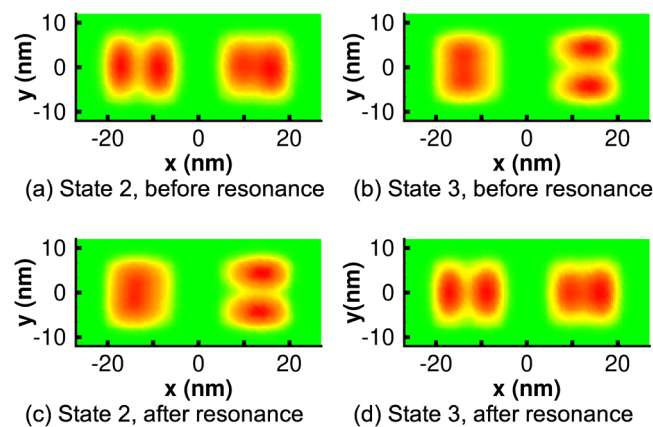


FIG. 2. Electron densities in a double QD with asymmetric sizes *before* the resonance R_{23} (with $w_2 = 14.76$ nm) are plotted in (a) for the second excited state and in (b) for the third excited state, while the electron densities *after* R_{23} (with $w_2 = 14.82$ nm) are plotted for the second excited state in (c) and for the third state in (d). The size of the first QD is fixed at $w_1 = 15$ nm. The distance between the QDs is $d = 10$ nm. Note that (a) and (b) evolve into (d) and (c), respectively.

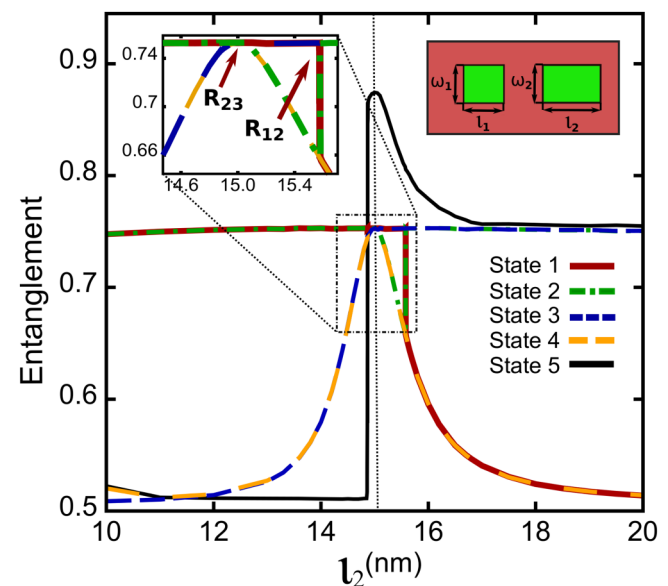


FIG. 3. Spatial entanglement of the first few excited eigenstates of two electrons in a double QD with non-symmetric lengths are plotted as a function of the length of the second QD ℓ_2 . The width of the first QD is fixed at $w_1 = 15$ nm. The distance between the QDs is $d = 10$ nm.

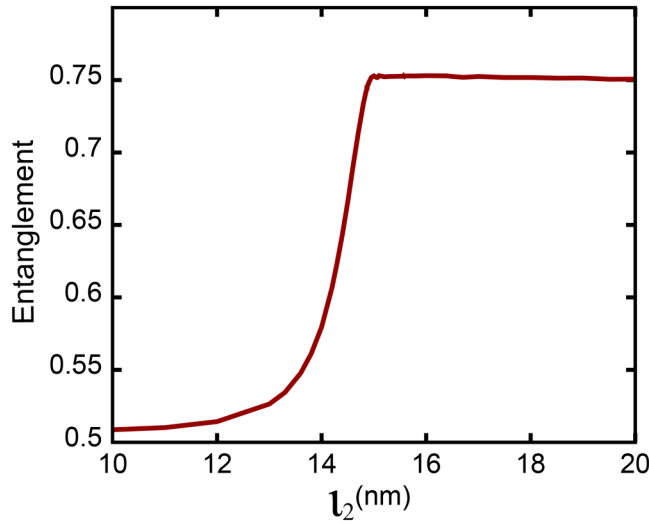


FIG. 4. Spatial entanglement of the third excited eigenstates of two electrons in a double QD of non-symmetric lengths is plotted as a function of the length of the second QD l_2 . The width of the first QD is fixed at $w_1 = 15$ nm. The distance between the QDs is $d = 10$ nm. The stepwise nature of the curve allows for the designing of a stable single-level entanglement QD system.

not change the nature of the geometry. In 2D and in higher dimensions, on the other hand, we enable the manipulation of the shapes more readily and thereby observe novel outcomes.

C. The double quantum dot with an applied B-field

In this subsection, we study the effect of an external magnetic field on electron entanglement. We consider a 2D symmetric double QD system under the influence of a constant magnetic field $\mathbf{B}$ directed perpendicular to the QD plane. The perturbing Hamiltonian is given by

$$H' = \frac{e}{m} \mathbf{A} \cdot \mathbf{p} + \frac{e^2}{2m} |\mathbf{A}|^2 - \boldsymbol{\mu} \cdot \mathbf{B}, \quad (5)$$

where $\boldsymbol{\mu} = -(e/m)\mathbf{S}$ with $\mathbf{S}$ being the spin angular momentum. The vector potential $\mathbf{A}$ is given in the Landau gauge to be $\mathbf{A} = \hat{y} Bx$. The spin-orbit coupling term does not exert any change on the spatial wavefunction, and therefore is not important to evaluating the spatial entanglement. With the inclusion of the magnetic field, the system has the additional effective potential terms xy and x^2 , with the former giving rise to a breaking of mirror symmetry in the potential profile. The entanglement of the system for a few excited states as a function of $\mathbf{B}$ are plotted in Fig. 5.

As seen from Fig. 5, the perturbation created by the magnetic field gives rise to dips in entanglement values of electron states in the system. At these dips, the single-particle state in each QD is assigned unambiguously. An examination of the electron densities confirms this claim. In Figs. 6(b) and 6(e), the electron densities corresponding to the lowest point of the entanglement values for

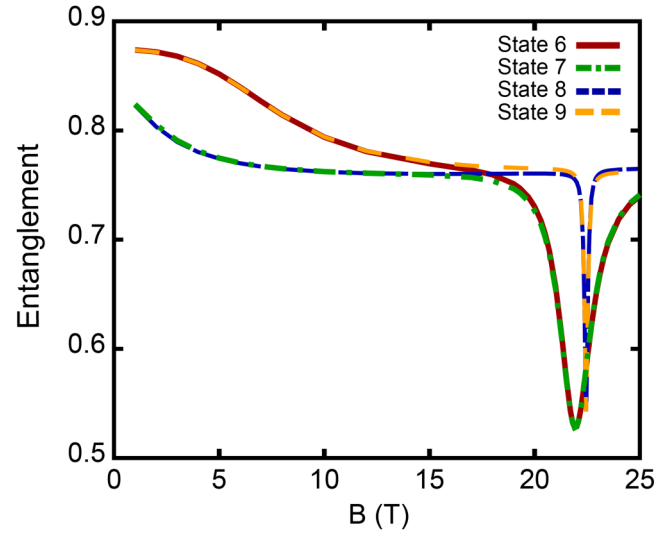


FIG. 5. Spatial entanglement of a few excited states of two electrons in a double QD with an external magnetic field $\mathbf{B}$ applied perpendicularly to the QD plane. The sizes of the QDs are fixed at $w_1 = w_2 = 15$ nm. The distance between the QDs is $d = 10$ nm.

the sixth and seventh states are plotted; we observe distinct patterns in the two QDs. However, in Figs. 6(a), 6(c), 6(d), and 6(f), we see that the electron densities at the QDs are no longer distinct when the system moves away from the entanglement dips.

D. The double quantum dot with impurities

In this section, we investigate how impurities degrade the level of entanglement in quantum dot systems. Consider a double square quantum dot with both QDs having the same size, $w_1 = w_2 = 15$ nm, with a distance $d = 10$ nm between them. In one of the two QDs, we introduce a circular impurity potential, modeled by a Gaussian distribution. It is given by

$$V_{im}(\mathbf{r}) = V_{io} \exp\left(-\frac{|\mathbf{r} - \mathbf{r}_{q2}|^2}{2\delta^2}\right), \quad (6)$$

where $\mathbf{r}_{q2}$ is the center of QD2 where the impurity is placed and δ^2 is the variance of the impurity potential distribution.

First, we study how the size of this impurity affects the entanglement. In Fig. 7, the dependence of entanglement on the characteristic radius δ is plotted. The amplitude V_{io} is set to 20 meV. We observe that except for the ground state whose entanglement is firmly set at 0.5, the next four excited states have their entanglement reduced as the size of the impurity increases. We also observe a level crossing between the second and third excited state (see Fig. 7). The reduction of entanglement as well as the crossing of states is consequences of the impurity breaking the symmetry of the double QD system.

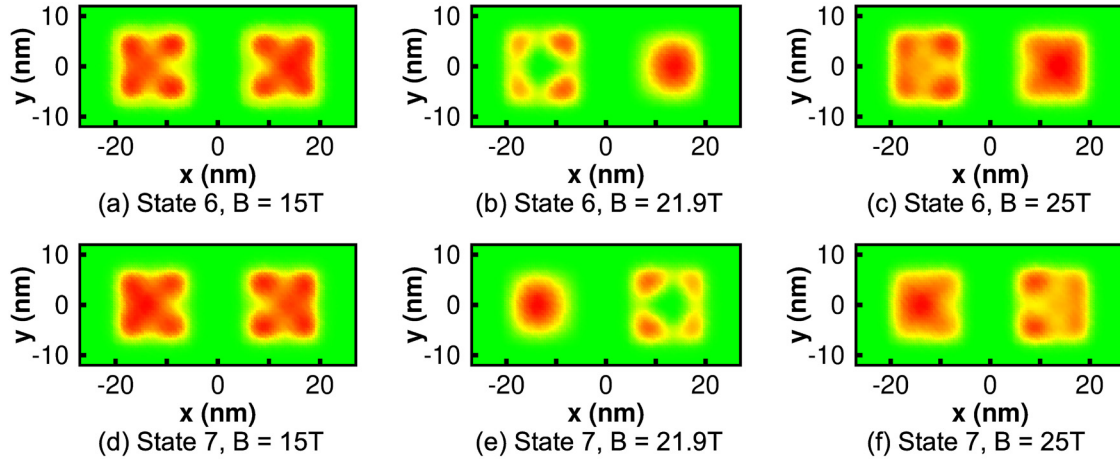


FIG. 6. Electron densities in a double QD subject to an external magnetic field B (a) before the dip in entanglement, (b) at the dip in entanglement, and (c) after the dip in entanglement as the applied magnetic field increases.

Next, we study how the amplitude of the potential impurity affects the entanglement. In Fig. 8, the entanglement of the first excited state is plotted as a function of δ for different values of $V_{io} = \{20, 40, -20, -40 \text{ meV}\}$. We observe that as δ increases, the drop in the entanglement when $V_{io} = \pm 40 \text{ meV}$ is larger than when $V_{io} = \pm 20 \text{ meV}$. This is because a higher value of $|V_{io}|$ corresponds to stronger asymmetry, which leads to lower entanglement. Therefore, this observation agrees with examples of

asymmetric systems discussed in Secs. III A–III C. The second observation is that in both the cases of $|V_{io}| = 20 \text{ meV}$ and $|V_{io}| = 40 \text{ meV}$, the positive impurity potential results in higher entanglement than negative impurity does.

Finally, we study the case of multiple impurities in the system. Consider the double square QD with QD2 containing n number of equally sized impurities, as shown in Fig. 9 for $n = \{1, 2, 3, 4\}$. The distance between two adjacent impurities is

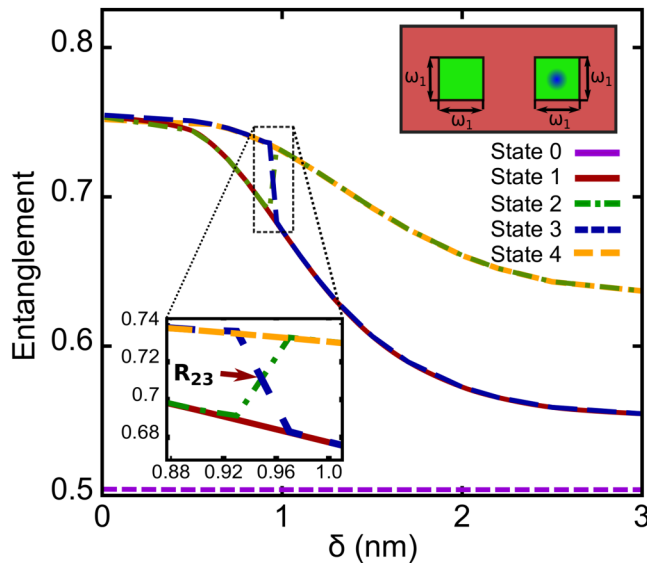


FIG. 7. Spatial entanglement of a few eigenstates of two electrons in a double QD with an impurity are plotted as a function of the characteristic radius δ of the impurity. The amplitude of the impurity is $V_{io} = 20 \text{ meV}$.

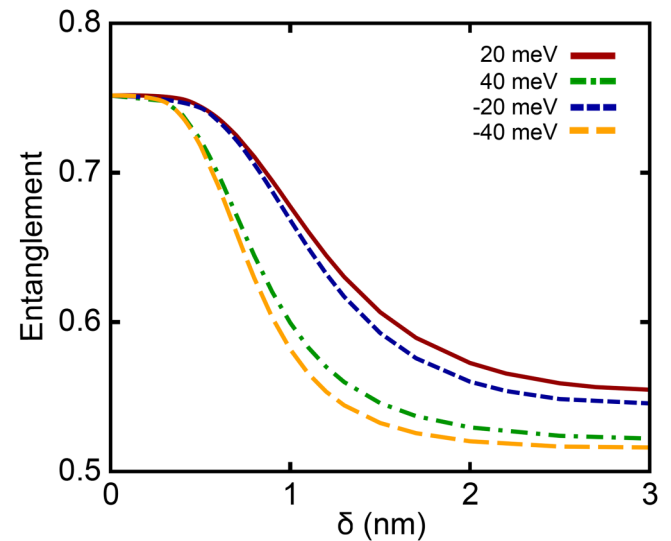


FIG. 8. Spatial entanglement of the first excited state of two electrons in a double QD with a single impurity is plotted as a function of the radius δ , with different values for the amplitude V_{io} of the impurity.

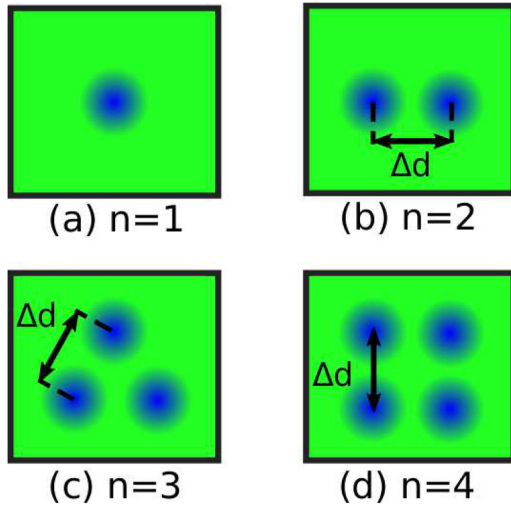


FIG. 9. Schematics of a quantum dot with (a) one impurity, (b) two impurities, (c) three impurities, and (d) four impurities. The impurities in a quantum dot are spaced so that any two adjacent impurities are separated by the same distance Δd .

$\Delta d = 5$ nm, with the amplitude $V_{io} = 20$ meV, and the characteristic radius is $\delta = 1$ nm. In Fig. 10, where the discrete plot of entanglement as a function of n is shown, we observe a level crossing between the second and third excited states; the entanglement level for the two states cross each other as the system

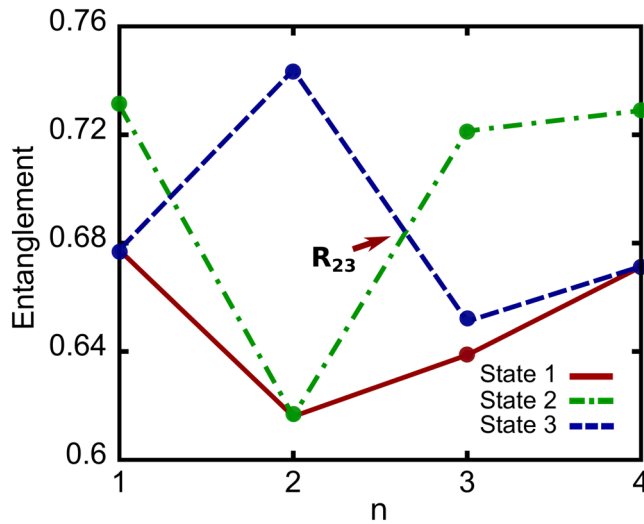


FIG. 10. Spatial entanglement of two electrons in a double QD is plotted as a function of the number of impurities n in one of the two QDs. The arrangement of impurities in the QD follows Fig. 9. The distance between two adjacent impurities is $\Delta d = 5$ nm, with amplitude $V_{io} = 20$ meV, and the characteristic radius is $\delta = 1$ nm.

evolves from two impurities to three impurities. This crossing is analogous to the crossing between the same states in Fig. 7, which is facilitated by creating asymmetry through an increase in the radius of single impurity.

E. Double quantum dots with a linking bridge

In many quantum dot systems, it is common to have tunneling bridges connecting the quantum dots to couple the electrons. In this subsection, we study one such scenario, where two QDs in a double square QD system are linked together via a bridge of width h having the same potential depth as the QD (see the system schematic in Fig. 11). The dependence of entanglement on the width h of the bridge is shown in Fig. 11. The size of both QDs is $w_1 = w_2 = 15$ nm, and the distance between the QDs is fixed at 10 nm. Since the system has mirror symmetry there is no reduction of entanglement from a high value to 0.5 as observed in systems with broken symmetry in Secs. III A–III D. Regardless, as shown in Fig. 11, we observe multiple level crossings between adjacent states as the width of the bridge expands.

The electron densities of the fifth and sixth excited states before and after their shared level crossing (labeled R_{56} in Fig. 11) is shown in Fig. 12. The electron density before the crossing is plotted for $h = 3$ nm, and the density after the states cross is plotted at $h = 3.2$ nm. Also from Fig. 12, we see that with the inclusion of the potential bridge, the electron density is slightly distorted as the tunnel draws a certain amount of electron distribution into it. This attraction is restrained by the Coulomb repulsion between the two electrons that forces them to be localized in their

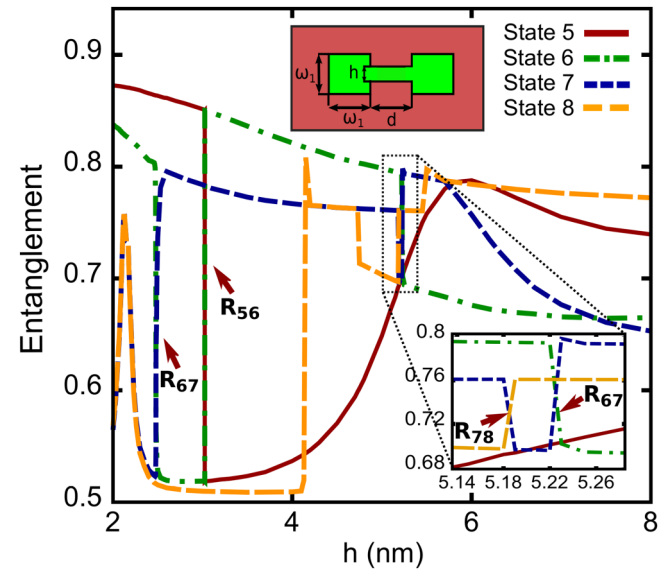


FIG. 11. Spatial entanglement of a few eigenstates of two electrons in a double QD connected through a tunneling bridge are plotted as a function of width h of the bridge.

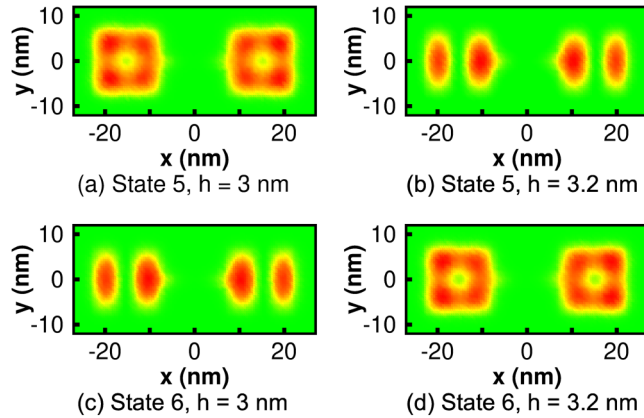


FIG. 12. Electron densities in a double QD with a tunneling bridge *before* the resonance R_{56} (with $h = 3$ nm) are plotted in (a) for the fifth excited state and in (b) for the sixth excited state, while the electron densities *after* R_{56} (with $h = 3.2$ nm) are plotted for the fifth excited state in (c) and for the sixth state in (d). The sizes of the QDs are fixed at $w_1 = w_2 = 15$ nm. The distance between the QDs is $d = 10$ nm.

respective QDs and prevents the electron distribution from occupying the entire bridge.

As the width h of the bridge increases, the potential profile of the system becomes more distorted and the electrons are no longer localized in the quantum dots. The evolution in electron density of the sixth excited state from when the double QD are separated to when a tunnel of width $h = 8$ nm is included, is shown in Fig. 13. In Fig. 13(b), we observe the appearance of electron distribution at the middle of the bridge as a result of h being large enough to draw the electrons away from their quantum dots. This is when a cluster of two electrons in the ground state has lower energy than when the electrons are far away from each other (hence having less Coulomb energy) but oscillate with high energies at excited states. Therefore, the electron cluster in the ground state, although it has a higher Coulomb potential, has less energy overall and is energetically preferred.

The distortion in electron density also results in a change of entanglement, which is shown in Fig. 11. At $h = 8$ nm, the amount of entanglement in the sixth state is not close to any known

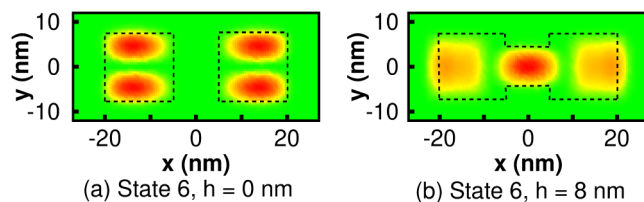


FIG. 13. Electron density of the sixth excited state in a double QD with no tunneling bridge is plotted in (a), while the electron density for when the tunnel has a width $h = 8$ nm is plotted in (b). The sizes of the QDs are fixed at $w_1 = w_2 = 15$ nm. The distance between the QDs is $d = 10$ nm.

asymptotic entanglement constants since the system is in the middle of transitioning from a one-electron-per-dot state to an electron cluster state.

F. Double triangular quantum dots

So far in the article, we have investigated the properties of spatial entanglement in different variations of the double square QDs. Although we have limited our numerical study to specific geometries, the observed phenomena and conclusions are generalizable to double QDs of arbitrary shapes. To further demonstrate the versatility of the numerical approach introduced in this article, we report the results on triangular double QDs. Although the analytical solutions for a single electron trapped in quantum dots of simpler shapes (equilateral triangular QDs in 2D³⁰ and tetrahedral QDs in 3D³¹) are known, the problems of two interacting electrons

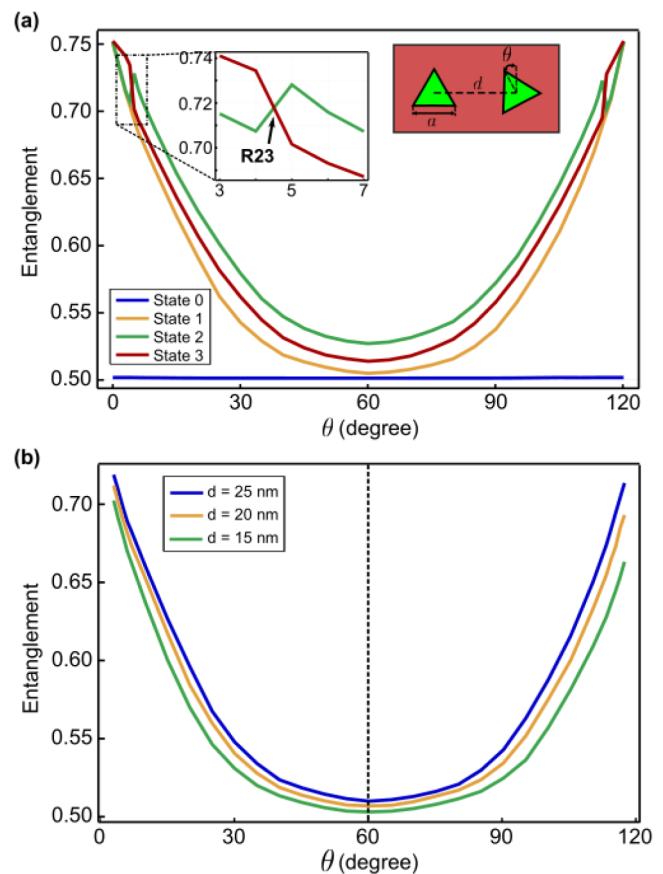


FIG. 14. Spatial entanglement of double triangular QD is plotted as a function of the relative angle θ between the orientations of the two QDs (specific definition of θ is shown in the inset). The QDs are equilateral triangles with sides $a = 12$ nm. (a) The entanglement values of the first few states of the system are plotted with the distance between the QDs being fixed at $d = 25$ nm. (b) The entanglement of the first excited state is plotted for different values of the distance d .

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in a double QDs of these shapes placed in the vicinity of each other can only be solved numerically and hence serve as additional demonstrations of our method. In particular, we are interested in how the relative orientation of one QD with respect to the other affects the spatial entanglement of the two-electron system. Consider two QDs, both are equilateral triangles with sides $a = 12$ nm whose circumcenters are separated by a distance d [see the schematic in Fig. 14(a)]. The change in the spatial entanglement between two trapped electrons as the relative angle θ between bases of the two triangular QDs varies from 0 to 120° is shown in Fig. 14(a) for the first few eigenstates. Since an equilateral triangle returns to its original orientation for every 120° rotation, the entanglement value of a state is maximized at $\theta \in \{0, 120\}$ when the double QD system is mirror symmetric. This is similar to the observations found in Secs. III A–III E. For the value of θ in between these end points, entanglement is reduced and is the lowest at $\theta = 60^\circ$, when the two-QD system is most strongly perturbed from its mirror-symmetric form. As also shown in Fig. 14(a), there are level crossings (between second and third states) due to the breaking of mirror symmetry brought about by the finite rotation of one of the two QDs. One can also see that the entanglement plots are roughly symmetric across the 60° mark. This symmetry point occurs because when the two quantum dots are sufficiently far from each other so that they only weakly interact, a θ -rotation of one QD by either counterclockwise or clockwise introduces roughly the same perturbation to the wavefunction at that QD. As a result, the entanglement value at 120° is approximately the same as it is at 0° . As the two QDs are brought closer to each other and the resulting wavefunction at one QD is more significantly distorted due to the presence of the other, this approximate symmetry across 60° no longer holds, as can be seen in Fig. 14(b). The smaller d is, the more asymmetric the entanglement profile becomes.

The symmetry in spatial entanglement value is not unique to the rotated triangular quantum dots, but a feature exhibited by all

quantum dots that have discrete rotational symmetries. As shown in Fig. 15, where we plot the spatial entanglement of the first few states of two electrons trapped in square double QDs, where one of the two QDs is rotated by an angle θ with respect to the other QD. The entanglement values for states 1, 2, and 3 peak at $\theta \in \{0, 90\}$ and are mirror symmetric across the 45° mark, where the particles are least entangled. This is because a square QD returns to its original orientation after every 90° rotation.

IV. CONCLUDING REMARKS

In this paper, we have studied the spatial entanglement of electrons trapped in various configurations of two-dimensional quantum dots. We showed that (i) a system of two electrons in a double quantum dot, where each electron is in a different single-particle state, has the maximum entanglement if the QDs are identical to each other, (ii) the entanglement of such states collapses to near 0.5 if the two QDs have similar shapes but in different sizes, (iii) the evolution of entanglement when more asymmetry is introduced into the system has a mirror symmetry behavior, (iv) and we also reported resonances in entanglement associated with level crossings between adjacent states.

Through our study of double quantum dot systems with asymmetric length, we showed that the mirror symmetry, which was observed when two QDs are of similar shape, disappears; instead, we observe stepwise shifts in entanglement between the two values 0.5 and 0.75, which suggest the possibility of designing single-level entanglement systems using asymmetric double QDs.

Our study of the effect of the external magnetic field on electron entanglement shows that the perturbation in the overall potential profile creates dips in the level of entanglement at which the electron densities at each of the quantum dots are distinct from each other.

We studied how impurities in a quantum dot may affect the entanglement of the system. We showed that the amount of entanglement in each state reduces as the size and the amplitude of the impurity increases, with level crossings being prevalent. As the number of distinct impurities increases, the level crossings also exist in the discrete entanglement spectrum.

We also discussed the entanglement in a double quantum dot system with a potential tunnel that couples the two QDs. We observe multiple level crossings as the width of the potential bridge expands. Formations of electron clusters as the bridge becomes wide enough and draws the electrons toward its center is seen.

Using double triangular QDs and double square QDs as examples, we studied how the relative orientation of one QD to the other affects the spatial entanglement of the two-electron system. A reduction in entanglement measurement when one QD is rotated in relation to the other is observed, along with level crossings due to the breaking of mirror symmetry as rotation occurred. For two weakly interacting QDs, the entanglement profile is symmetric across the point where the difference in the orientations of the two QDs is largest, which is also where the entanglement is lowest.

Here, we have shown that when electrons are trapped inside QDs in proximity, the spatial correlation between them becomes significant and gives rise to a controllable spatial entanglement with many exciting features. Through our study of entanglement in

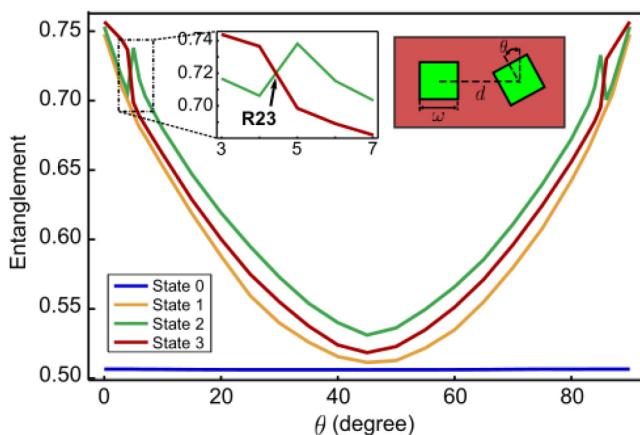


FIG. 15. Spatial entanglement of double square QD is plotted as a function of the relative angle θ between the orientations of the two QDs (see the inset). The QDs are identical squares with sides $\omega = 12$ nm. The distance between the QDs are fixed at $d = 25$ nm.

various quantum dot structures, we have shown that the level of entanglement can be precisely tuned by manipulation of the overall symmetry of the system, by varying the QD shape and smoothness, or by applying external fields. In addition to its tunability, the many interesting aspects that spatial entanglement displays unveil exciting possibilities of harnessing it for quantum information and quantum computing applications, as well as for the designing of electronic devices in the nanoscale regime.

While we have presented here the results for QDs of special shapes, we emphasize again that the method of studying entanglement is readily extended to QDs of any geometry, and the use of FEM allows us to transcend such geometrical aspects.

ACKNOWLEDGMENTS

We thank the Center for Computational NanoScience and the Academic and Research Computing Group at WPI for computational resources. We thank H. S. Mani for fruitful discussions.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Dung N. Pham: Conceptualization (equal); Formal analysis (equal); Methodology (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Sathwik Bharadwaj:** Formal analysis (equal); Writing – review & editing (equal). **L. R. Ram-Mohan:** Conceptualization (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

APPENDIX: THEOREMS ON THE ASYMPTOTIC VALUES OF SPATIAL ENTANGLEMENT IN DOUBLE QDS

In this Appendix, a brief presentation of the theorem on the asymptotic values of spatial entanglement is included here for completeness. We specifically discuss an analytical derivation for the spatial entanglement of two particles in symmetric double QDs when the QDs are asymptotically far from each other. First, let $|n, \alpha\rangle_i$ be the single-particle basis for the spatial wavefunctions in a single QD without the presence of the second QD. Here, n is the single-particle quantum number of the state, $\alpha \in \{1, 2\}$ is the index of the QD, and $i \in \{1, 2\}$ is the index of the particle. When the QDs are asymptotically far from each other such that their single-particle bound-state wavefunctions do not overlap, the two-particle spatial wavefunctions occupying the entire double QD system are

given by Ref. 23,

$$|\psi_s^+\rangle = \frac{1}{2\sqrt{1+\delta_{\alpha\beta}}} [|1, \alpha\rangle_1 |2, \beta\rangle_2 + |1, \beta\rangle_1 |2, \alpha\rangle_2 + |2, \alpha\rangle_1 |1, \beta\rangle_2 + |2, \beta\rangle_1 |1, \alpha\rangle_2], \quad (\text{A1})$$

$$|\psi_s^-\rangle = \frac{1}{2} [|1, \alpha\rangle_1 |2, \beta\rangle_2 - |1, \beta\rangle_1 |2, \alpha\rangle_2 - |2, \alpha\rangle_1 |1, \beta\rangle_2 + |2, \beta\rangle_1 |1, \alpha\rangle_2], \quad (\text{A2})$$

$$|\psi_a^+\rangle = \frac{1}{2} [|1, \alpha\rangle_1 |2, \beta\rangle_2 - |1, \beta\rangle_1 |2, \alpha\rangle_2 + |2, \alpha\rangle_1 |1, \beta\rangle_2 - |2, \beta\rangle_1 |1, \alpha\rangle_2], \quad (\text{A3})$$

$$|\psi_a^-\rangle = \frac{1}{2\sqrt{1+\delta_{\alpha\beta}}} [|1, \alpha\rangle_1 |2, \beta\rangle_2 + |1, \beta\rangle_1 |2, \alpha\rangle_2 - |2, \alpha\rangle_1 |1, \beta\rangle_2 - |2, \beta\rangle_1 |1, \alpha\rangle_2], \quad (\text{A4})$$

where to satisfy the antisymmetric condition of fermions, the symmetric spatial wavefunctions in Eqs. (A1) and (A2) are accompanied by the antisymmetric spin part, while the antisymmetric spatial wavefunctions (A3) and (A4) are to be coupled with the symmetric spin component. One can show that regardless of the specific spatial dependence of the QDs, as long as the double QD system is mirror symmetric, the spatial entanglement $\mathcal{E}_\ell$ of any of the four asymptotic wavefunctions above is equal to either 0.5 or 0.75. Here, we provide the proof for the wavefunction in Eq. (A1), as the derivations for the other three cases directly follow. The reduced density matrix for $|\psi_{sym}^1\rangle$ in Eq. (A1) is given by

$$\begin{aligned} \rho_1 &= \frac{1}{4(1+\delta_{\alpha\beta})} \int \langle \mathbf{r}_2 | \psi_{sym}^1 \rangle \langle \psi_{sym}^1 | \mathbf{r}_2 \rangle d\mathbf{r}_2 \\ &= \frac{1}{4(1+\delta_{\alpha\beta})} \int \langle \langle \mathbf{r}_2 | 2, \beta \rangle_2 | 1, \alpha \rangle_1 + \langle \mathbf{r}_2 | 2, \alpha \rangle_2 | 1, \beta \rangle_1 \\ &\quad + \langle \mathbf{r}_2 | 1, \beta \rangle_2 | 2, \alpha \rangle_1 + \langle \mathbf{r}_2 | 1, \alpha \rangle_2 | 2, \beta \rangle_1 \rangle \langle \langle 2, \beta | \mathbf{r}_2 \rangle_2 \langle 1, \alpha | \\ &\quad + \langle 2, \alpha | \mathbf{r}_2 \rangle_2 \langle 1, \beta | + \langle 1, \beta | \mathbf{r}_2 \rangle_2 \langle 2, \alpha | + \langle 1, \alpha | \mathbf{r}_2 \rangle_2 \langle 2, \beta | \rangle d\mathbf{r}_2 \\ &= \frac{1}{4(1+\delta_{\alpha\beta})} [|1, \alpha\rangle \langle 1, \alpha| + |1, \beta\rangle \langle 1, \beta| + |2, \alpha\rangle \langle 2, \beta| \\ &\quad + |2, \beta\rangle \langle 2, \beta| + \delta_{\alpha\beta} (|1, \alpha\rangle \langle 1, \beta| + |1, \beta\rangle \langle 1, \alpha| \\ &\quad + |2, \alpha\rangle \langle 2, \beta| + |2, \beta\rangle \langle 2, \alpha|)]. \end{aligned} \quad (\text{A5})$$

In the last line above, we have dropped all the particle indices, since the second particle has been integrated out and all the remaining bras and kets are that of the first particle. The spatial

entanglement is then given by

$$\begin{aligned}
 \mathcal{E}_\ell &= 1 - \text{Tr}(\rho_1^2), \\
 &= 1 - \int |\langle \mathbf{r}'_1 | \rho_1 | \mathbf{r}_1 \rangle|^2 d\mathbf{r}'_1 d\mathbf{r}_1 \\
 &= 1 - \frac{1}{16(1 + \delta_{\alpha\beta})^2} \int \left| \langle \mathbf{r}'_1 | 1, \alpha \rangle \langle 1, \alpha | \mathbf{r}_1 \rangle + \langle \mathbf{r}'_1 | 1, \beta \rangle \langle 1, \beta | \mathbf{r}_1 \rangle \right. \\
 &\quad + \langle \mathbf{r}'_1 | 2, \alpha \rangle \langle 2, \alpha | \mathbf{r}_1 \rangle + \langle \mathbf{r}'_1 | 2, \beta \rangle \langle 2, \beta | \mathbf{r}_1 \rangle \\
 &\quad + \delta_{\alpha\beta} (\langle \mathbf{r}'_1 | 1, \alpha \rangle \langle 1, \beta | \mathbf{r}_1 \rangle + \langle \mathbf{r}'_1 | 1, \beta \rangle \langle 1, \alpha | \mathbf{r}_1 \rangle \\
 &\quad \left. + \langle \mathbf{r}'_1 | 2, \alpha \rangle \langle 2, \beta | \mathbf{r}_1 \rangle + \langle \mathbf{r}'_1 | 2, \beta \rangle \langle 2, \alpha | \mathbf{r}_1 \rangle \right|^2 d\mathbf{r}'_1 d\mathbf{r}_1.
 \end{aligned} \tag{A6}$$

Given the orthonormality condition $\langle i, \alpha | j, \beta \rangle = \delta_{ij} \delta_{\alpha\beta}$, we see that the expression on the last line of Eq. (A6) reduces to

$$\mathcal{E}_\ell = \begin{cases} 0.5, & \alpha = \beta, \\ 0.75, & \alpha \neq \beta, \end{cases} \tag{A7}$$

which are precisely the values observed and discussed in the main text for when the two QDs are sufficiently far from each other.

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